

Day 4: Network Cable Types and Tools Theory

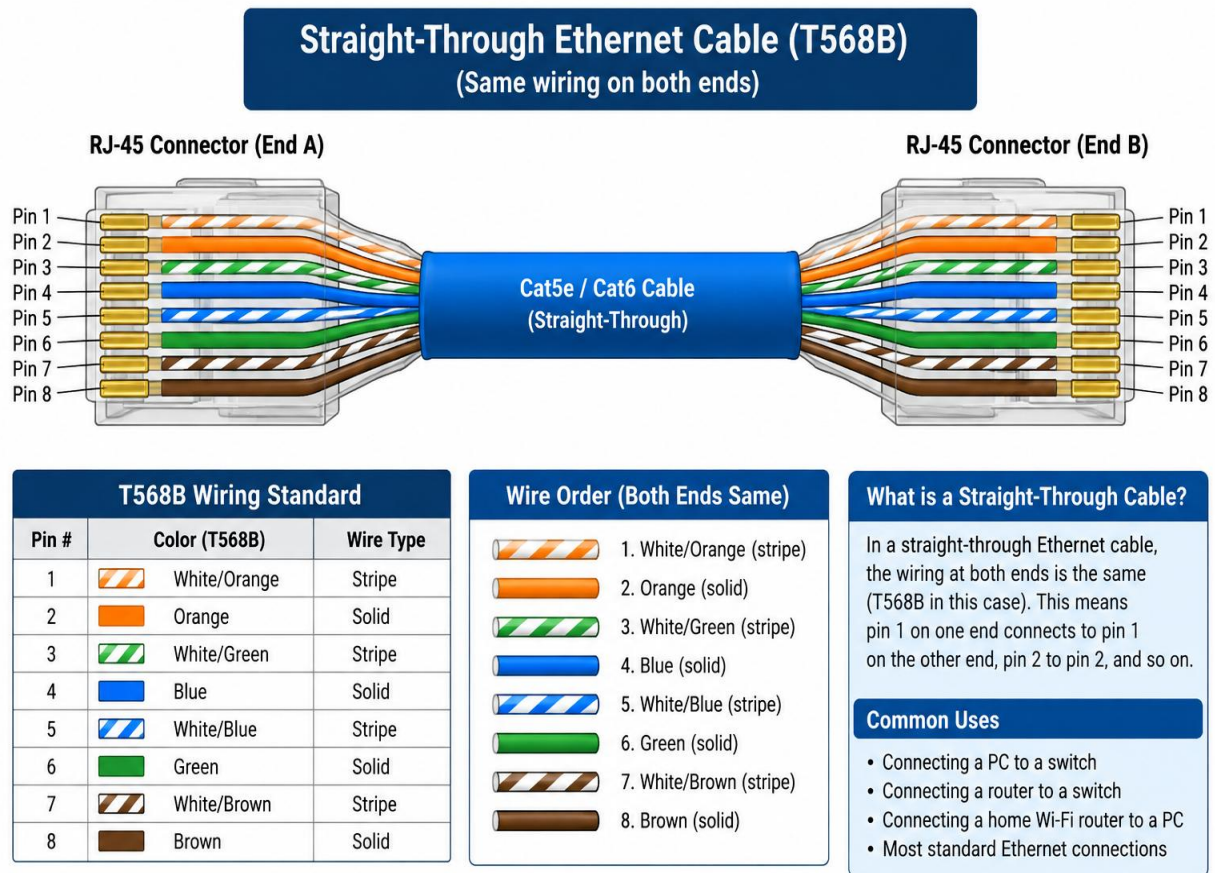
Task 1 : List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Feature	Ethernet Cable	Fiber Optic Cable
1. Transmission medium	Uses copper wires to transmit electrical signals.	Uses glass/plastic fibers to transmit light signals.
2. Speed & distance	Good speed, but generally shorter distances (typically up to 100 m for common copper Ethernet).	Supports very high speeds and much longer distances, from hundreds of meters to many kilometers depending on the fiber type.
3. Interference	Can be affected by electromagnetic interference (EMI) .	Immune to EMI , making it better for electrically noisy environments.

Real-world examples

- **Ethernet:** Connecting a **home Wi-Fi router to a PC or switch** using a Cat6 Ethernet cable.
- **Fiber Optic:** Connecting buildings across a **college campus backbone**, where long-distance, high-speed communication is required.

Task 2 : Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



Note: A straight-through Ethernet cable uses the same T568B wiring standard on both ends.

Task 3 : Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Steps to Attach an RJ45 Connector to a CAT6 Cable

Materials Required:

- CAT6 Ethernet cable
- RJ45 connector
- RJ45 crimping tool
- Cable stripper/cutter
- Cable tester (optional)

Step 1: Cut the Cable

Cut the CAT6 cable to the required length using a cable cutter.

Step 2: Strip the Outer Jacket

Use the stripping tool to carefully remove about 1–2 cm of the outer jacket from the end of the cable. Be careful not to damage the internal wires.

Step 3: Untwist the Wires

Separate the four twisted pairs and gently untwist them. Arrange the eight wires according to the **T568B standard**.

T568B Color Order:

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

Step 4: Straighten and Align the Wires

Straighten the wires with your fingers and arrange them neatly in the correct order. Trim the wire ends evenly so that they can enter the RJ45 connector properly.

Step 5: Insert the Wires into the RJ45 Connector

Hold the RJ45 plug with the clip facing down and slide all eight wires into the connector. Make sure each wire reaches the front of the connector and remains in the correct pin order.

Step 6: Check the Wiring

Before crimping, check that the wires are still in the correct T568B order and that all eight wires have reached the end of the connector.

Step 7: Crimp the Connector

Place the RJ45 connector into the crimping tool and squeeze the handles firmly. The crimping tool locks the connector onto the cable and pushes the contacts into the individual wires.

Step 8: Test the Cable

If a cable tester is available, connect both ends of the cable to the tester. Check that pins **1 through 8** are detected in the correct order.

Result:

The CAT6 cable is now terminated with an RJ45 connector and can be used for an Ethernet network connection.

Task 4 : Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

- For a gaming cafe with 10 PCs, I would choose **Cat6 Ethernet cables** instead of fiber optic cables.
- Cat6 Ethernet provides enough speed and low latency for LAN gaming, and the PCs are usually placed within a relatively short distance of the network switch.
- Ethernet cables are also more **affordable and easier to install**, which makes them a better choice for a small gaming cafe with a limited budget.
- I would use an **RJ45 crimper** to attach RJ45 connectors to the Cat6 cables, while a fiber cleaver would only be needed if I were installing fiber optic cables.